

2016.0 RANGE ROVER (LG), 501-00

BODY SYSTEM - GENERAL INFORMATION

THREAD REPAIR INSERT (G1585192)

GENERAL PROCEDURES

SPECIAL TOOL(S)



**JLR-501-
169**

Steel Insert Kit



**JLR-501-
173**

M20 Steel Insert
Kit



E166127

JLR-501-175

Extractor, Steel Insert

REPAIR



M18 steal inserts

CAUTIONS:

- Do not use air tools while carrying out this procedure.
- If one hole/thread is damaged, all remaining holes/threads for the affected subframe, must have a M18 steel insert installed.
- This procedure requires the aid of another technician to monitor the alignment of the drill in both axis. Failure to follow this instruction may result in damage to the body and incorrect subframe tighten torque.
- The drill bits within the special tool kit must only be used for this repair. If the drill bits are used for non aluminium use, a new special tool kit must be used.
- Make sure that the drill speed is not at maximum (**maximum drill speed does not exceed 300 RPM**). Use a suitable electric drill with a slow speed setting.

NOTE:

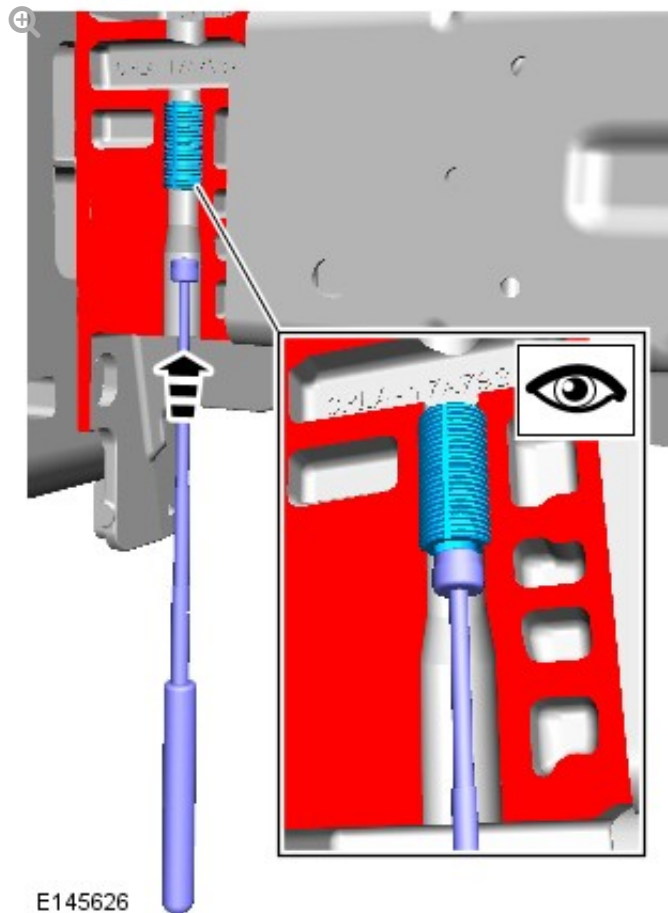
Some variation in the illustrations may occur, but the essential information is always correct.

1. **Steps 2-14 must be carried out for the M18 inserts. Do not carry out any Steps from Step 16 onwards for this size of insert.**

2.

WARNING:

Prior to carrying out this procedure, make sure that the chassis is not already installed with an insert. Using a suitable extended magnet, check the damaged hole for a steel insert. Do not drill if there is a steel insert located already. Failure to follow this instruction may result in permanent damage to the vehicle.



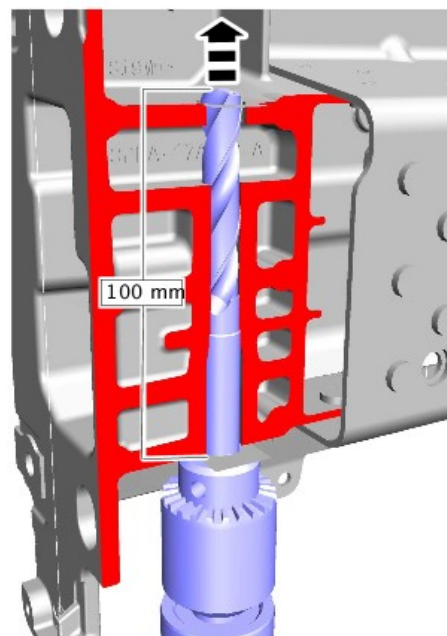
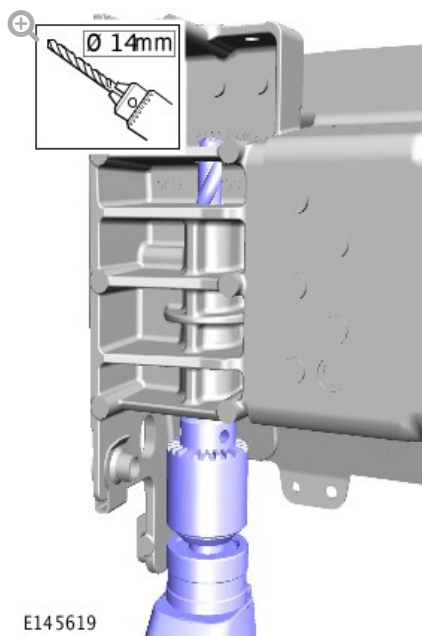
Using compressed air and suitable cleaning solution, make sure the hole is clean and free of debris.

CAUTIONS:

- **Make sure that the 14mm drill is used on the first stage of this procedure. Failure to follow this instruction may result in further damage to the vehicle.**
- Make sure that the depth of the drill does not exceed 100mm.

NOTES:

- **Apply a suitable cutting lubricate to the drill bit prior to drilling.**
- Measure 100mm from the top of the drill and apply suitable tape from the bottom of the measurement to the use as a depth gauge.



Drill the hole to a depth of 100mm using a 14mm drill.

4.

CAUTION:

Make sure that the inside of the drilled hole is clean and free of oil residue.

Using compressed air and suitable cleaning solution, make sure the hole is clean and free of debris.

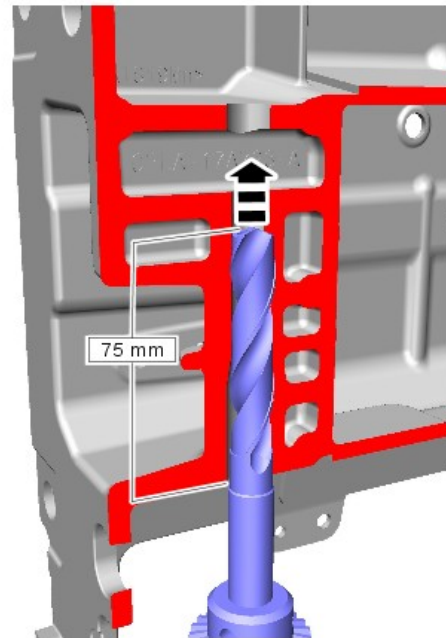
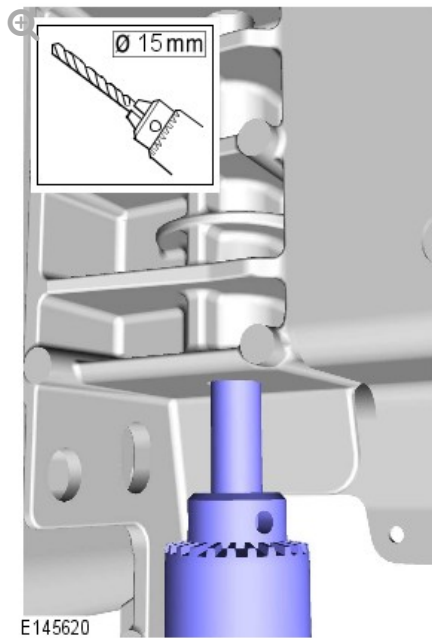
5.

CAUTION:

Make sure that the depth of the drill does not exceed 75mm.

NOTES:

- **Apply a suitable cutting lubricate to the drill bit prior to drilling.**
- Measure 75mm from the top of the drill and apply suitable tape from the bottom of the measurement to the use as a depth gauge.



Drill the hole to a depth of 75mm using a 15mm drill.

6.

CAUTION:

Make sure that the inside of the drilled hole is clean and free of oil residue.

Using compressed air and suitable cleaning solution, make sure the hole is clean and free of debris.

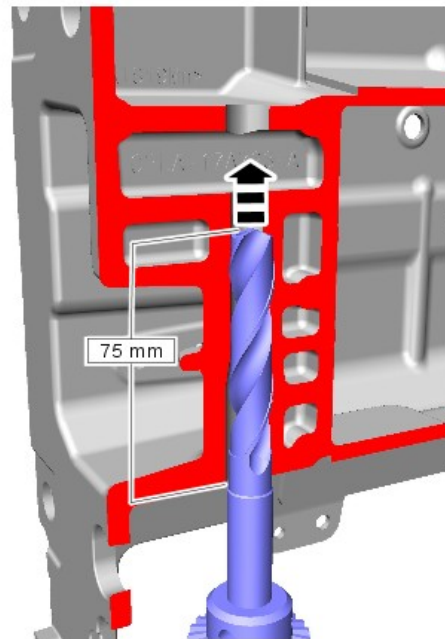
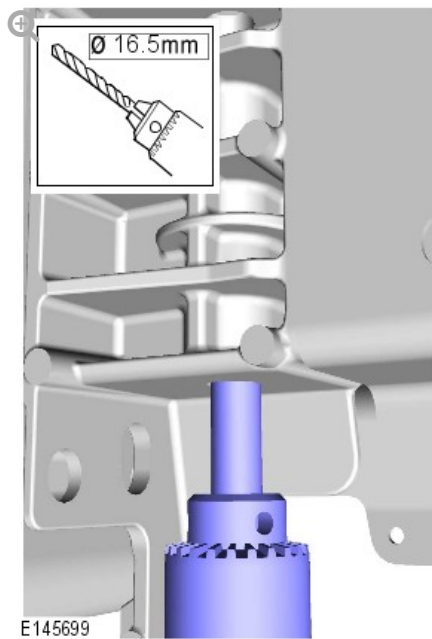
7.

CAUTION:

Make sure that the depth of the drill does not exceed 75 mm

NOTES:

- **Apply a suitable cutting lubricate to the drill bit prior to drilling.**
- Measure 75mm from the top of the drill and apply suitable tape from the bottom of the measurement to the use as a depth gauge.



Drill the hole to a depth of 75mm using a 16.5mm drill.

8.

CAUTION:

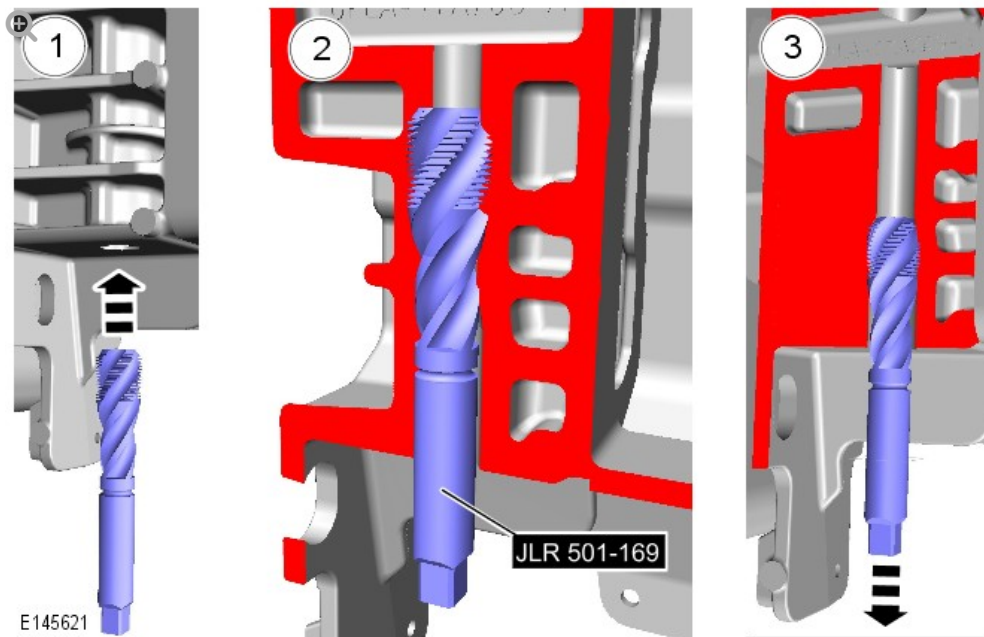
Make sure that the inside of the drilled hole is clean and free of oil residue.

Using compressed air and suitable cleaning solution, make sure the hole is clean and free of debris.

9.

NOTES:

- **Apply a suitable cutting lubricate to the tap prior to cutting the thread.**
- Tap the hole at half a turn increments and then back off one eighth of a turn. This must be carried out for each cut of the tap.



Tap the hole using a M18 x 1.5 flute tap.

Special Tool(s): [JLR-501-169](#)

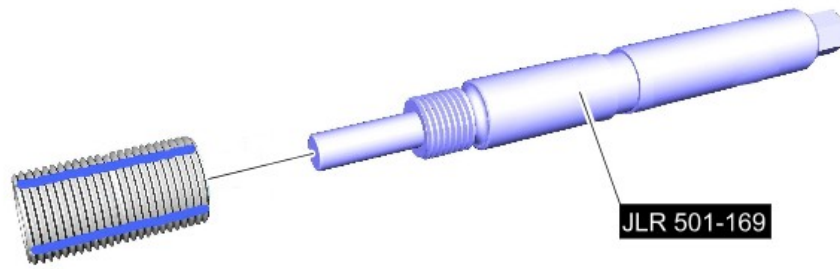
10.

CAUTION:

Make sure that the inside of the drilled hole is clean and free of oil residue.

Using compressed air and suitable cleaning solution, make sure the hole is clean and free of debris.

11.



E149222

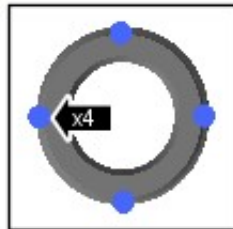
Thread the steel insert on to the special tool.

Special Tool(s): [JLR-501-169](#)

12.

CAUTION:

Make sure that the threaded steel insert is clean and free of oil residue before applying Loctite.



E145720

Apply four beads of Loctite 243 to the outside of the threaded steel insert as shown.

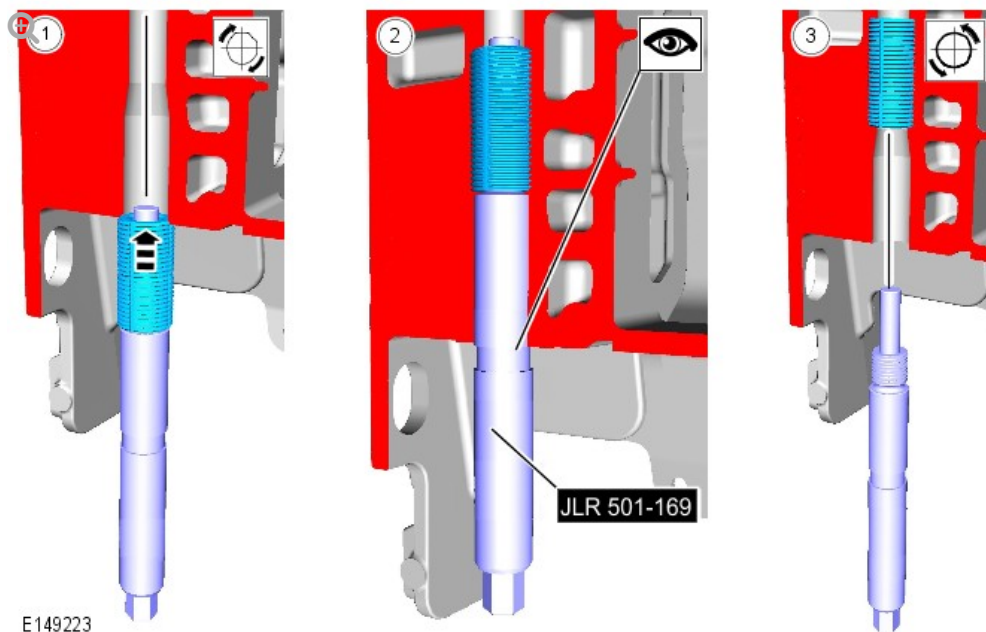
13.

CAUTION:

Make sure that the upper machined cut on the special tool is flush with the flat surface.

NOTE:

Before removing the special tool, use a suitable hammer and gently strike the special tool to release it from the threads. This will release the steel insert from the special tool and allow the special tool to be removed.



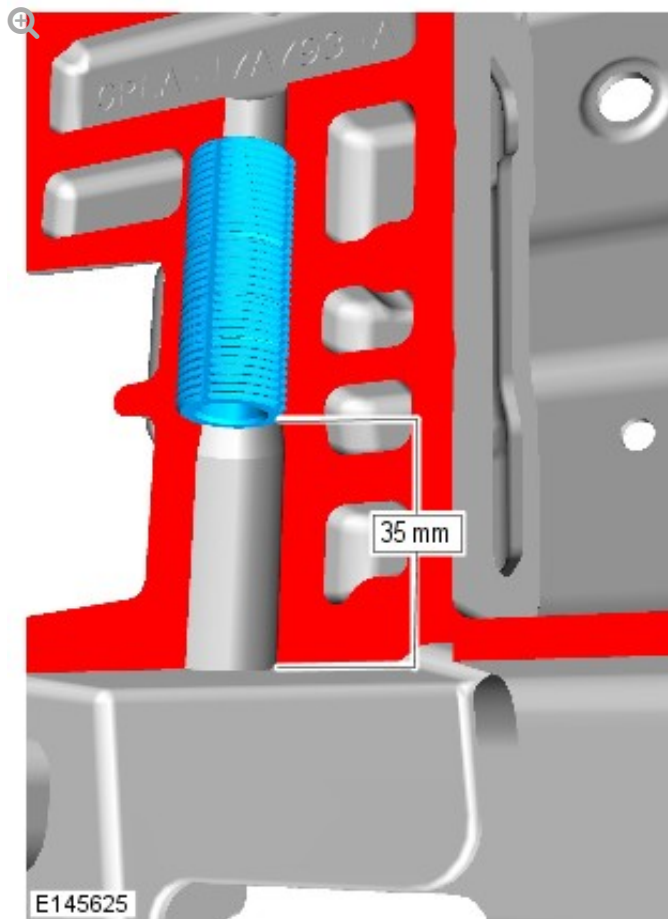
Using the special tool, install the threaded insert into the body.
Note the position of the machined cut on the special tool in relation to the body.

Special Tool(s): [JLR-501-169](#)

14.

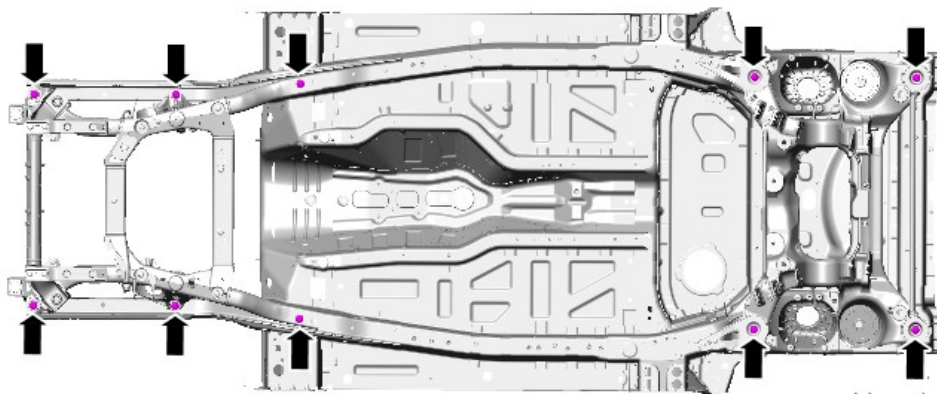
CAUTION:

Allow the Loctite to cure (30 minutes) before installing any of the subframe retaining bolts.



Using a suitable tool measure the seated depth of the steel insert, if the depth isn't as shown repeat step 14.

15.



E149241

Carry out Steps 2-14 for the remaining subframe fixings.



CAUTIONS:

- Do not use air tools while carrying out this procedure.
- This procedure requires the aid of another technician to monitor the alignment of the drill in both axis. Failure to follow this instruction may result in damage to the body and incorrect subframe tighten torque.
- The drill bits within the special tool kit must only be used for this repair. If the drill bits are used for non aluminium use, a new special tool kit must be used.
- Make sure that the drill speed is not at maximum (**maximum drill speed does not exceed 300 RPM**). Use a suitable electric drill with a slow speed setting.

NOTE:

Some variation in the illustrations may occur, but the essential information is always correct.

16.

CAUTION:

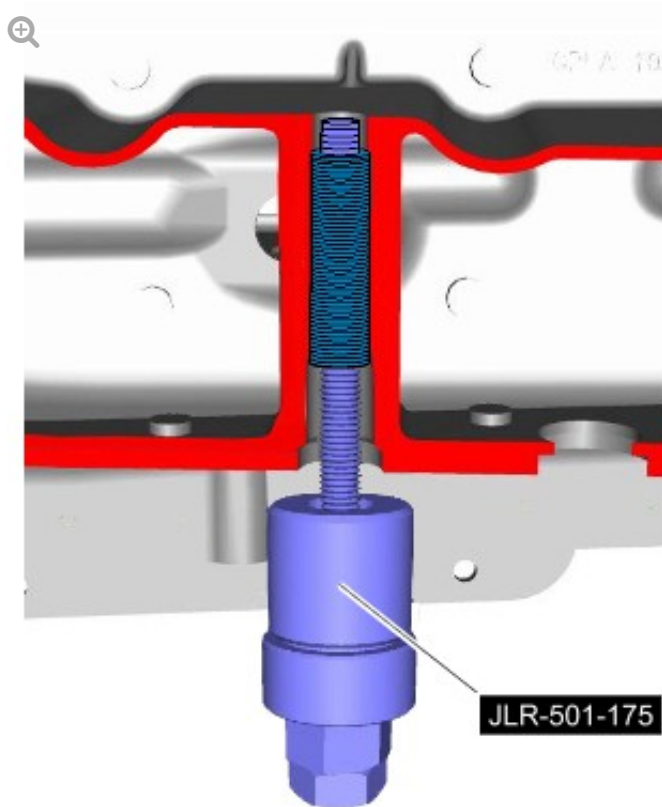
If the M18 steel insert cannot hold the torque applied during installation of the subframe retaining bolts, new M20 steel inserts must be installed.

NOTE:

Only install a M20 steel insert to an affected hole/thread. Not all holes/threads require the M20 steel insert if they have not failed.

Steps 17-29 must be carried out for the M20 inserts.

17.

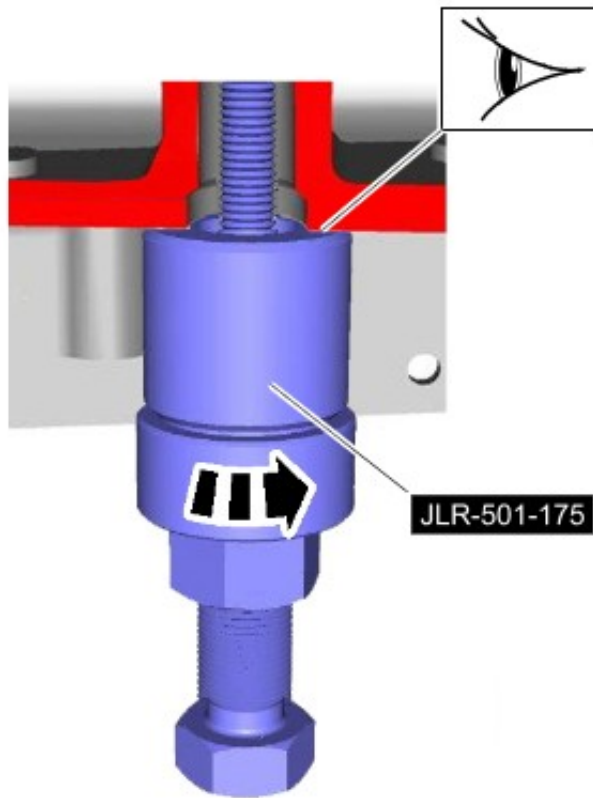


E172922

Install the special tool in to the steel insert until it cannot go any further.

Special Tool(s): [JLR-501-175](#)

18.



E172923

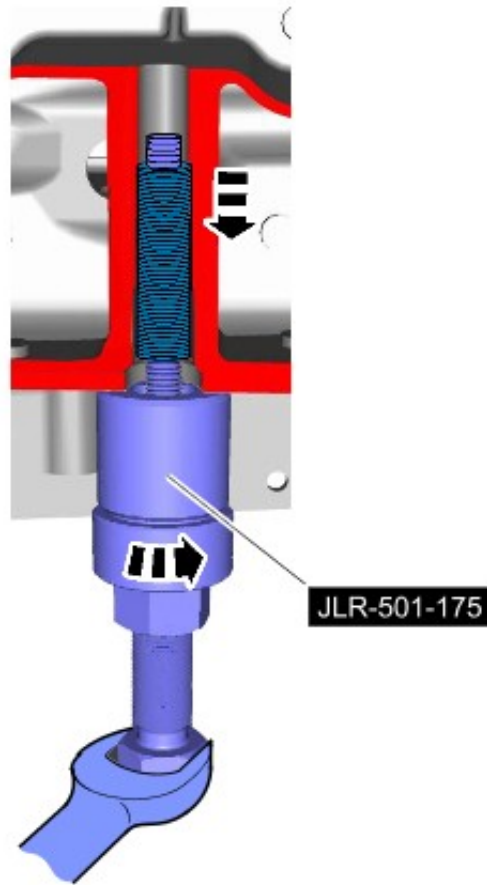
Rotate the locking ring of the special tool clockwise until the special tool is flush to the body and cannot rotate any further.

Special Tool(s): [JLR-501-175](#)

19.

NOTE:

Make sure that the internal shaft of the special tool does not rotate during this operation.



E172924

Rotate the locking ring of the special tool clockwise and extract the steel insert.

Special Tool(s): [JLR-501-175](#)

20.

CAUTION:

Make sure that the inside of the drilled hole is clean and free of oil residue.

Using compressed air and suitable cleaning solution, make sure the hole is clean and free of debris.

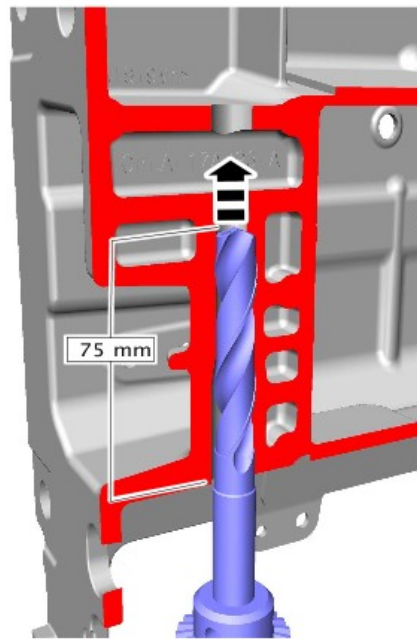
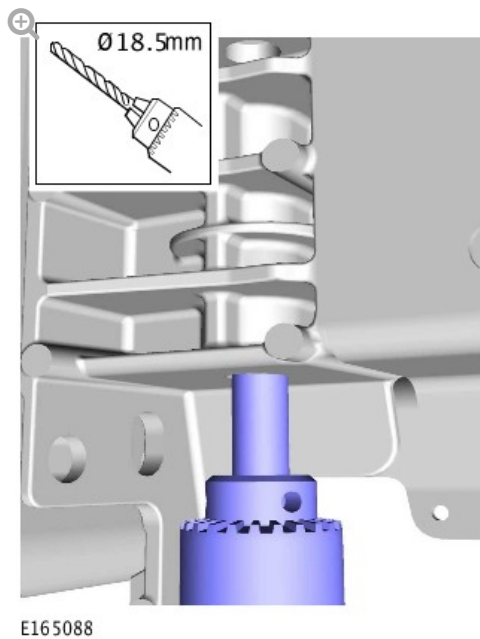
21.

CAUTION:

Make sure that the depth of the drill does not exceed 75mm.

NOTES:

- Apply a suitable cutting lubricate to the drill bit prior to drilling.
- Measure 75mm from the top of the drill and apply suitable tape from the bottom of the measurement to the use as a depth gauge.



Drill the hole to a depth of 75mm using a 18.5mm drill.

Special Tool(s): [JLR-501-173](#)

22.

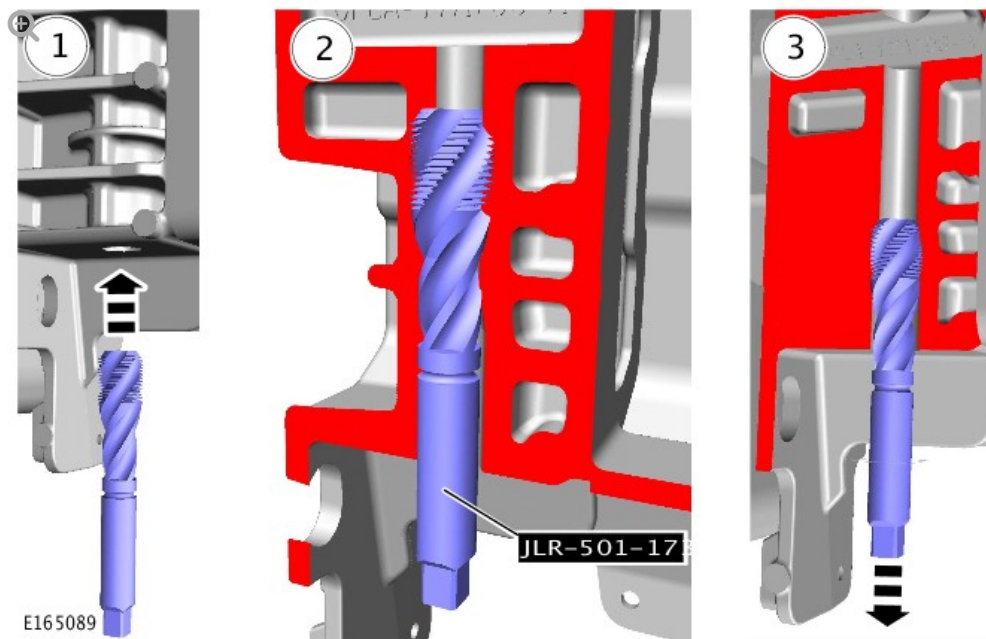
CAUTION:

Make sure that the inside of the drilled hole is clean and free of oil residue.

Using compressed air and suitable cleaning solution, make sure the hole is clean and free of debris.

NOTES:

- **Apply a suitable cutting lubricate to the tap prior to cutting the thread.**
- Tap the hole at half a turn increments and then back off one eighth of a turn. This must be carried out for each cut of the tap.



Tap the hole using a M20 x 1.5 flute tap.

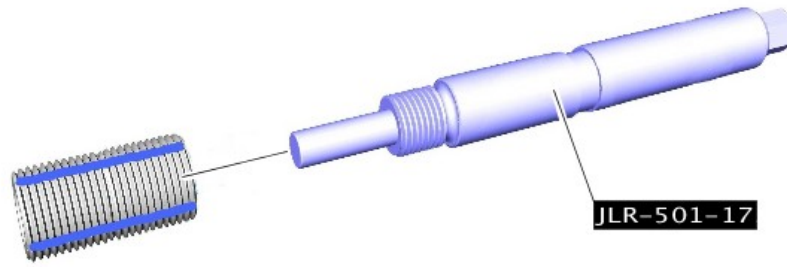
Special Tool(s): [JLR-501-173](#)

CAUTION:

Make sure that the inside of the drilled hole is clean and free of oil residue.

Using compressed air and suitable cleaning solution, make sure the hole is clean and free of debris.

25.



E165090

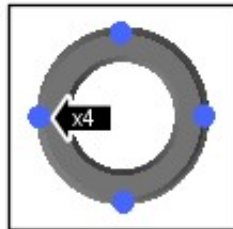
Thread the steel insert on to the special tool.

Special Tool(s): [JLR-501-173](#)

26.

CAUTION:

Make sure that the threaded steel insert is clean and free of oil residue before applying Loctite.



E145720

Apply four beads of Loctite 243 to the outside of the threaded steel insert as shown.

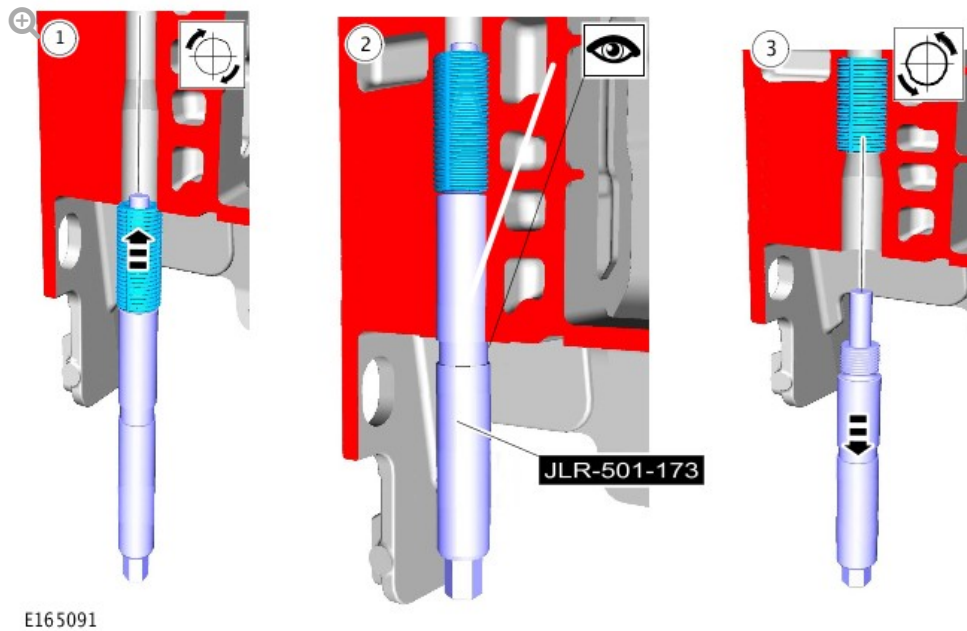
27.

CAUTION:

Make sure that the upper machined cut on the special tool is flush with the flat surface.

NOTE:

Before removing the special tool, use a suitable hammer and gently strike the special tool to release it from the threads. This will release the steel insert from the special tool and allow the special tool to be removed.



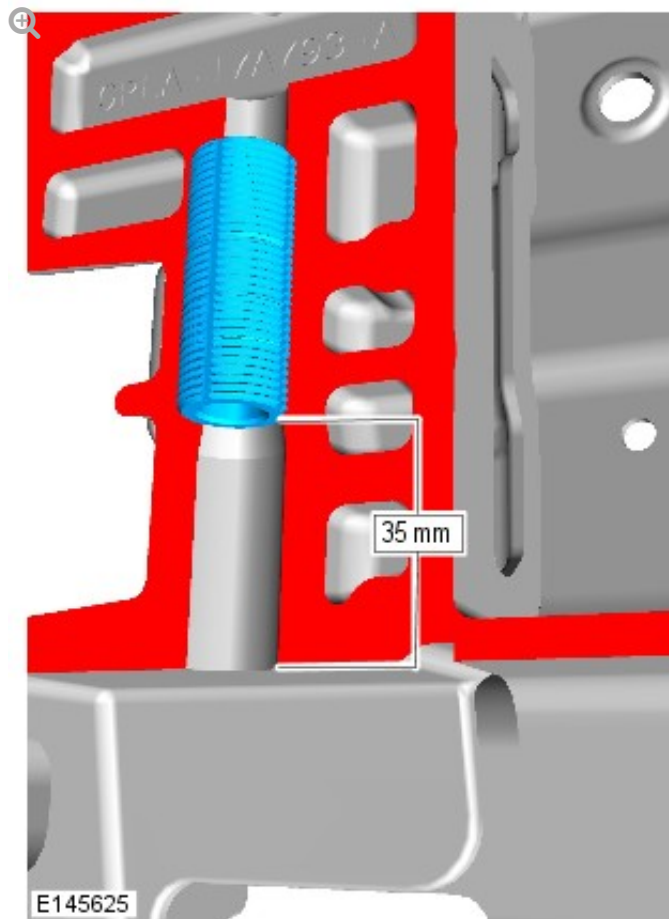
Using the special tool, install the threaded insert into the body.
Note the position of the machined cut on the special tool in relation to the body.

Special Tool(s): [JLR-501-173](#)

28.

CAUTION:

Allow the Loctite to cure (30 minutes) before installing any of the subframe retaining bolts.



Using a suitable tool measure the seated depth of the steel insert, if the depth isn't as shown repeat Step 26.